

Problem Statement: LearnMate – Agentic AI for Personalized Course Pathways

The Challenge

Students often struggle to identify the right learning path due to the overwhelming number of online courses and the lack of personalized guidance. Learning resources are scattered across platforms, making it difficult to choose relevant courses and build structured career-focused roadmaps.

The Objective

Build an AI-powered personalized learning assistant that recommends courses based on interests, skills, and career goals. The solution should provide adaptive learning pathways, skill assessment, career guidance, progress tracking, and Retrieval-Augmented Generation (RAG)-based knowledge retrieval using IBM Granite Models and IBM watsonx.ai on IBM Cloud Lite.

Key Features

- Personalized course recommendations.
- Dynamic learning roadmaps.
- Skill gap analysis and career guidance.
- Progress tracking dashboard.
- Multi-agent AI architecture with RAG.